

Sprint 1 - AI Module Tasks

Working document for planning the first iteration of the AI module in the adaptive e-learning platform.

The goal of Sprint 1 is to deliver a minimal end-to-end flow: the system receives a request, reads the student history, computes a simple mastery score, selects an appropriate question, and avoids previously seen questions.

Task	Objective	Final Outcome
Task 1	AI Endpoint	Backend can call the AI service
Task 2	Student History Retrieval	AI has real data for decisions
Task 3	Simple Mastery Calculation	Produces a score between 0 and 1
Task 4	Question Selection	Chooses the optimal question
Task 5	Seen Questions Filtering	Avoids repetition

Task 1 - AI Endpoint

Objective

Expose an endpoint so the backend can request a new question.

Implementation

- Define serializer for input validation.
- Implement view to handle requests.
- Create route /recommend/.
- Return JSON response with question data.

How Objective is Achieved

Achieved when backend receives valid responses from AI.

What the Task Delivers

Delivers the external interface of the AI module.

Definition of Done

- POST endpoint works.
- Invalid data is handled.
- Valid JSON response returned.
- Logic delegated to engine.

Task 2 - Student History Retrieval

Objective

Allow AI to access student history and candidate questions.

Implementation

- Define models for questions and interactions.
- Implement repository methods for history.
- Fetch seen questions.
- Fetch candidate questions.

How Objective is Achieved

Achieved when AI can retrieve all necessary data.

What the Task Delivers

Provides data layer for AI decisions.

Definition of Done

- Models exist.
- Repository returns required data.
- Seed/test data available.
- Data usable by engine.

Task 3 - Simple Mastery Calculation

Objective

Transform student history into a mastery score.

Implementation

- Compute accuracy and average time.
- Create DTO for features.
- Implement rule-based estimator.
- Return mastery score.

How Objective is Achieved

Achieved when score reflects student performance.

What the Task Delivers

Delivers first intelligence layer.

Definition of Done

- Features computed correctly.
- Score between 0-1.
- Test cases exist.
- Formula explainable.

Task 4 - Question Selection

Objective

Select question based on target difficulty.

Implementation

- Compute target difficulty.
- Select closest question.
- Use DTO for result.
- No generation yet.

How Objective is Achieved

Achieved when system selects appropriate question.

What the Task Delivers

Delivers adaptive recommendation.

Definition of Done

- Target computed.
- Correct topic used.
- Closest question selected.
- Fallback exists.

Task 5 - Seen Questions Filtering

Objective

Avoid recommending already seen questions.

Implementation

- Retrieve seen question IDs.

- Filter candidates.

- Define exclusion rule.

- Fallback if all seen.

How Objective is Achieved

Achieved when repetition is avoided.

What the Task Delivers

Improves user experience.

Definition of Done

- Seen questions excluded.

- Fallback defined.

- No repetition in demo.

- Works in flow.